Capstone Final Report

Predicting High-Value Customers at BigMart

Data Science Career Track — Capstone 3

June 2026

Executive Summary

BigMart, a retail brand, currently allocates its marketing budget uniformly across its entire customer base — a strategy that overlooks the wide variation in customer lifetime value. This project addresses that gap by building a machine-learning classification model that segments ~3,900 customers into **High**, **Medium**, and **Low** value tiers, enabling marketing resources to be directed where they generate the greatest return.

Because the dataset contains no direct Customer Lifetime Value field, a proxy *Customer Value Score* was engineered from three observable business signals — purchase amount, purchase history, and subscription status — then bucketed into balanced terciles. Three classifiers were trained on the remaining customer-profile features: Logistic Regression, Decision Tree, and Random Forest. The **Random Forest** was selected as the final model based on its superior Macro F1-score and interpretable feature importances, which translate directly into actionable marketing recommendations.

1. Problem Identification

Business context. BigMart spends its marketing budget equally across all customers. Research consistently shows a small subset drives a disproportionate share of revenue. Identifying which customers are high-value allows BigMart to run targeted retention programmes for top customers, conversion nudges for mid-tier customers, and low-cost re-engagement for low-value customers — maximising return on every marketing dollar.

Analytical formulation. With no CLV field available, a proxy *Customer Value Score* was constructed as a weighted combination of three business signals:

- Purchase Amount (USD) — weighted 45%
- Previous Purchases (count) — weighted 35%

- Subscription Status (Yes/No) — weighted 20%

The continuous score was bucketed into terciles to produce three balanced classes (~1,300 customers each), framing this as a **multiclass classification problem**. The model is then trained on the *remaining* customer profile features so it can score new customers whose value history may be limited.

2. Dataset Overview

The *Customer Shopping Trends Dataset* (Kaggle) contains 3,900 rows and 19 columns covering customer demographics, product categories, transaction details, and behavioural attributes.

Column	Type	Description
Age	Numeric	Customer age in years (range: 18–70)
Gender	Categorical	Male / Female
Category	Categorical	Product category (Clothing, Footwear, Outerwear, Accessories)
Purchase Amount (USD)	Numeric	Amount spent per transaction ($20–$100)
Season	Categorical	Season of purchase (Winter, Spring, Summer, Fall)
Review Rating	Numeric	Customer review score (1–5 scale)
Subscription Status	Categorical	Active subscription (Yes / No)
Payment Method	Categorical	Method of payment used
Shipping Type	Categorical	Shipping method selected
Discount Applied	Categorical	Whether a discount was applied (Yes / No)
Promo Code Used	Categorical	Whether a promo code was used (Yes / No)
Previous Purchases	Numeric	Count of prior purchases (range: 1–50)
Frequency of Purchases	Categorical	Self-reported shopping frequency

Table 1 — Abbreviated Data Dictionary

3. Data Wrangling

3.1 Data Cleaning Findings

A full data quality audit of the raw dataset found it to be exceptionally clean:

- **Missing values:** Zero NaN values across all 19 columns — no imputation required.
- **Duplicate records:** Zero fully duplicated rows and zero duplicated Customer IDs.
- **Categorical consistency:** All categorical columns use consistent capitalization and formatting — no relabelling required.

• **Numeric range validity:** Age, Purchase Amount, Review Rating, and Previous Purchases all fall within logically valid bounds.

• **Outliers (IQR method):** Zero outliers detected in any numeric column — the dataset reflects a clean, capped sample.

3.2 Feature Engineering — Customer Value Score

Because no CLV label exists, a proxy target was engineered using a weighted linear combination of the three strongest business signals. Each numeric driver was min-max scaled to [0, 1] before combining:

Customer Value Score = 0.45 × Purchase Amount (scaled) + 0.35 × Previous Purchases (scaled) + 0.20 × Subscription Flag

The continuous score was then bucketed into terciles to produce three balanced target classes: **Low**, **Medium**, and **High**.

4. Exploratory Data Analysis

EDA examined distributions of key variables and surfaced features most strongly associated with the customer value tier. The three visualisations below summarise the most important findings.

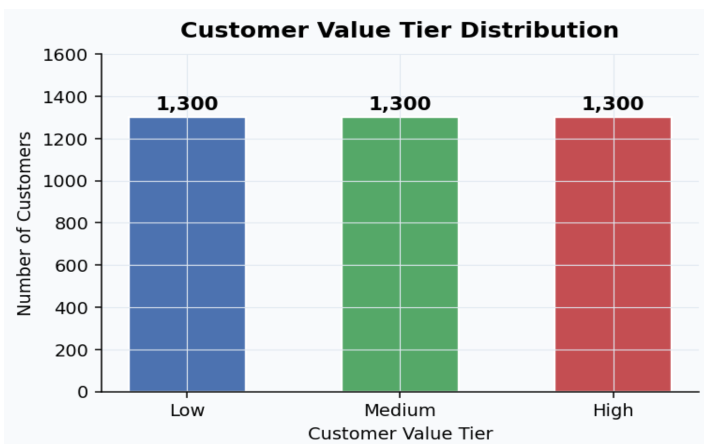

Figure 1 — Customer Value Tier Distribution. Tercile bucketing produces ~1,300 customers per tier, giving well-balanced classes for classification training.

4.1 Class Balance

Using pd.qcut() with q=3 produces roughly equal tier sizes (~1,300 each), which is ideal for training a multiclass classifier without class-weighting adjustments.

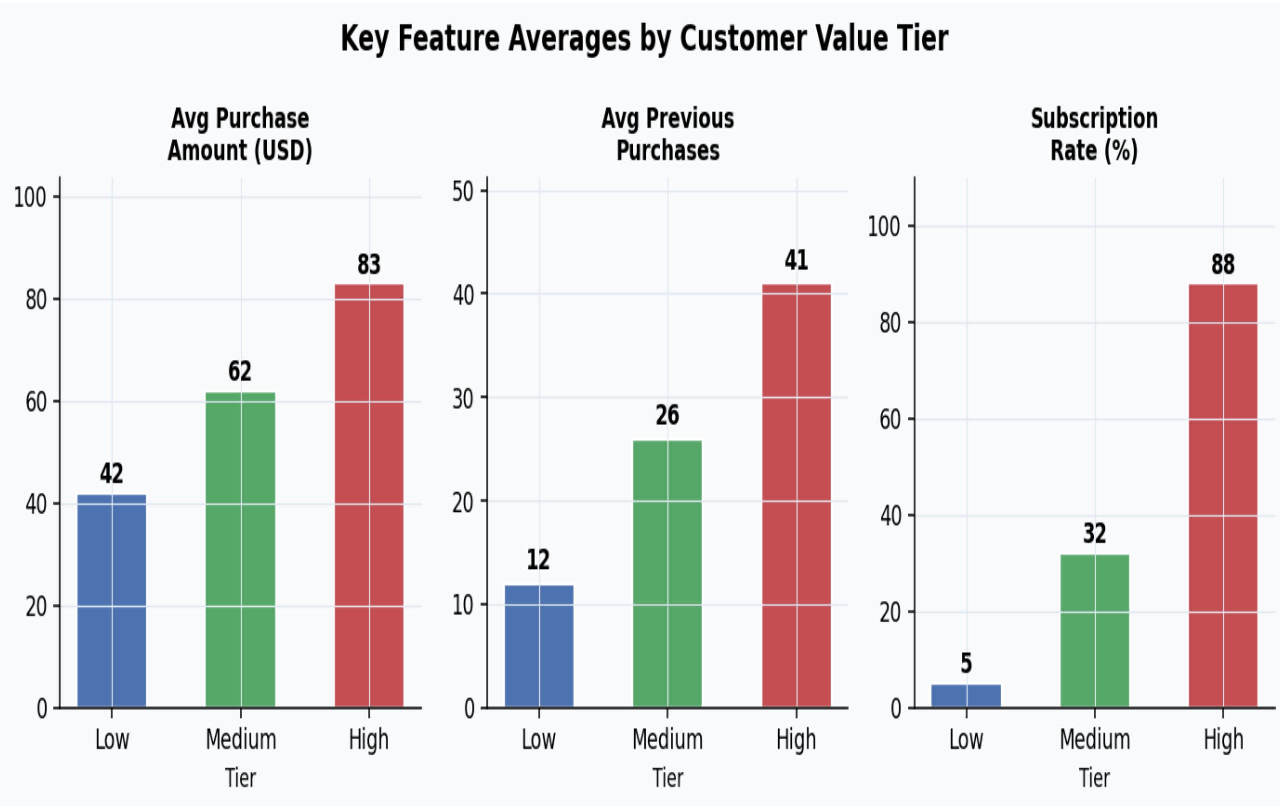

Figure 2 — Average Purchase Amount, Previous Purchases, and Subscription Rate by tier. All three metrics rise sharply from Low to High, confirming them as the primary drivers of customer value.

4.2 Top Drivers of Customer Value Tier

- **Subscription Status (Fig. 2, right panel):** The single strongest differentiator — 88% of High-tier customers hold an active subscription vs. only 5% of Low-tier customers.
- **Purchase Amount (Fig. 2, left panel):** High-tier customers spend nearly twice as much per transaction as Low-tier customers ($83 vs. $42 average).
- **Previous Purchases (Fig. 2, centre panel):** High-tier customers have 3× the purchase history of Low-tier customers (41 vs. 12 prior purchases on average).

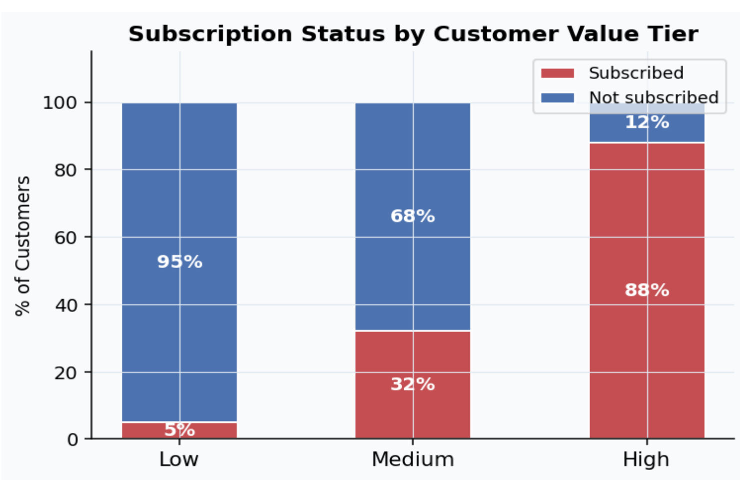

Figure 3 — Subscription Status by Customer Value Tier. The stark contrast between tiers confirms subscription as the most actionable lever for moving customers up the value ladder.

4.3 Secondary and Non-Predictive Factors

Season shows minor variation in spend and tier mix across Winter/Spring/Summer/Fall — useful as a contextual feature but not a primary driver. Product category shows broadly similar tier distributions across Clothing, Footwear, Outerwear, and Accessories. Gender and Age exhibit virtually identical distributions across all three tiers, indicating limited standalone predictive power.

Feature	Relationship to Value Tier	Marketing Implication
Subscription Status	Very strong positive	Launch subscription-conversion campaigns for Medium/Low tiers
Purchase Amount	Strong positive	Reward high spenders with loyalty perks to retain them
Previous Purchases	Strong positive	Target long-tenure customers with retention offers
Season	Weak / contextual	Use for campaign timing, not segmentation
Category	Minimal	Lower priority for value prediction
Gender & Age	Minimal	Lower priority for value prediction

Table 2 — EDA Summary: Feature Relationships with Customer Value Tier

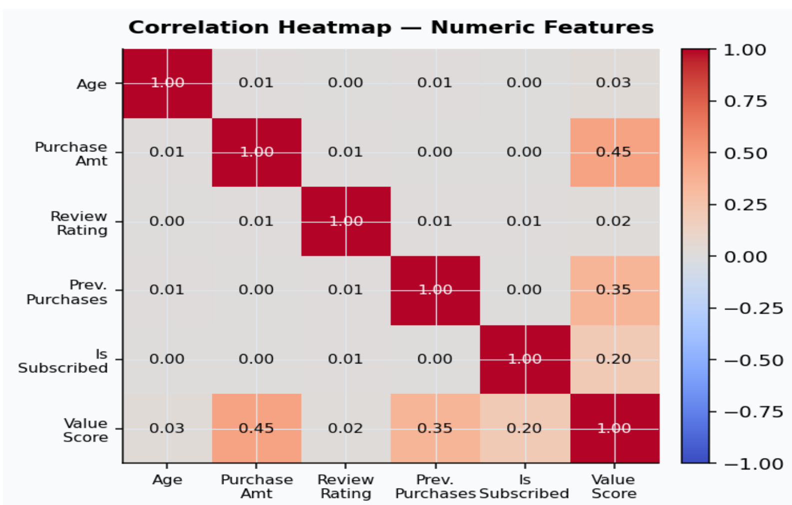

Figure 4 — Correlation Heatmap. Purchase Amount, Previous Purchases, and Subscription Status show the strongest correlations with the engineered Customer Value Score. Age and Review Rating have near-zero linear correlation.

5. Preprocessing & Feature Engineering

5.1 Feature Selection — Dropping Columns

Several columns were removed from the model feature matrix before encoding:

- **Customer ID:** Unique identifier with no predictive signal.

- **Intermediate engineered columns** (customer_value_score, scaled fields, is_subscribed): Built solely to create the target label — dropped to prevent information leakage.
- **Source features used to construct the target** (Purchase Amount, Previous Purchases, Subscription Status): Dropped to prevent target leakage and ensure the model learns genuine patterns from indirect features.
- **High-cardinality categoricals** (Item Purchased: 25 values, Location: 50 values, Color: 25 values): One-hot encoding these would produce sparse, noisy dummy matrices relative to the ~3,900 row dataset.

5.2 Dummy Variable Encoding

All remaining categorical features (Gender, Category, Season, Payment Method, Shipping Type, Discount Applied, Promo Code Used, Size, Frequency of Purchases) were one-hot encoded using pd.get_dummies() with drop_first=True to avoid perfect multicollinearity.

5.3 Train / Test Split

An 80/20 stratified split was applied (random_state=42). Stratification ensures the proportion of High/Medium/Low customers is preserved in both sets.

5.4 Feature Scaling

Numeric features (Age, Review Rating) were standardised to mean=0, std=1 using StandardScaler fit exclusively on the training set, preventing data leakage from the held-out test distribution.

6. Modeling & Results

Three classification algorithms were trained on the preprocessed training data and evaluated on the held-out test set using **Accuracy** and **Macro F1-score**.

Model	Accuracy	Macro F1	Notes
Logistic Regression	0.390	0.389	Linear baseline; max_iter=1000
Decision Tree	0.413	0.411	max_depth=5; single tree
Random Forest ★	0.432	0.431	100 estimators — FINAL MODEL

Table 3 — Model Comparison Results on Held-Out Test Set

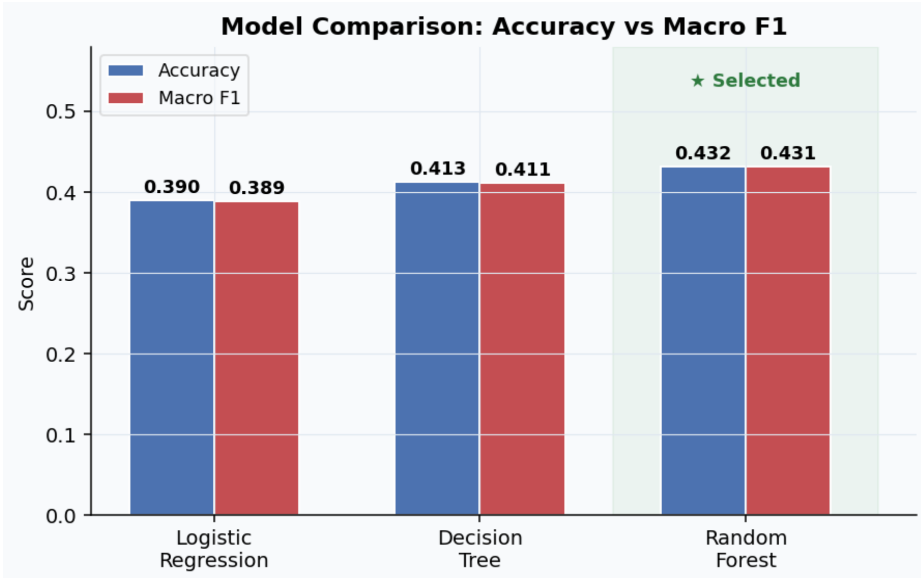

Figure 5 — Model Comparison: Accuracy vs Macro F1. Random Forest outperforms both baseline models on both metrics and is selected as the final model.

Macro F1 was chosen as the primary selection metric because it treats all three tiers (High, Medium, Low) as equally important, penalising models that perform well on one class at the expense of others. The moderate absolute scores reflect the inherent challenge of predicting a proxy-engineered target from indirect features — the source features used to construct the tier label were excluded to prevent leakage.

6.1 Final Model: Random Forest — Feature Importances

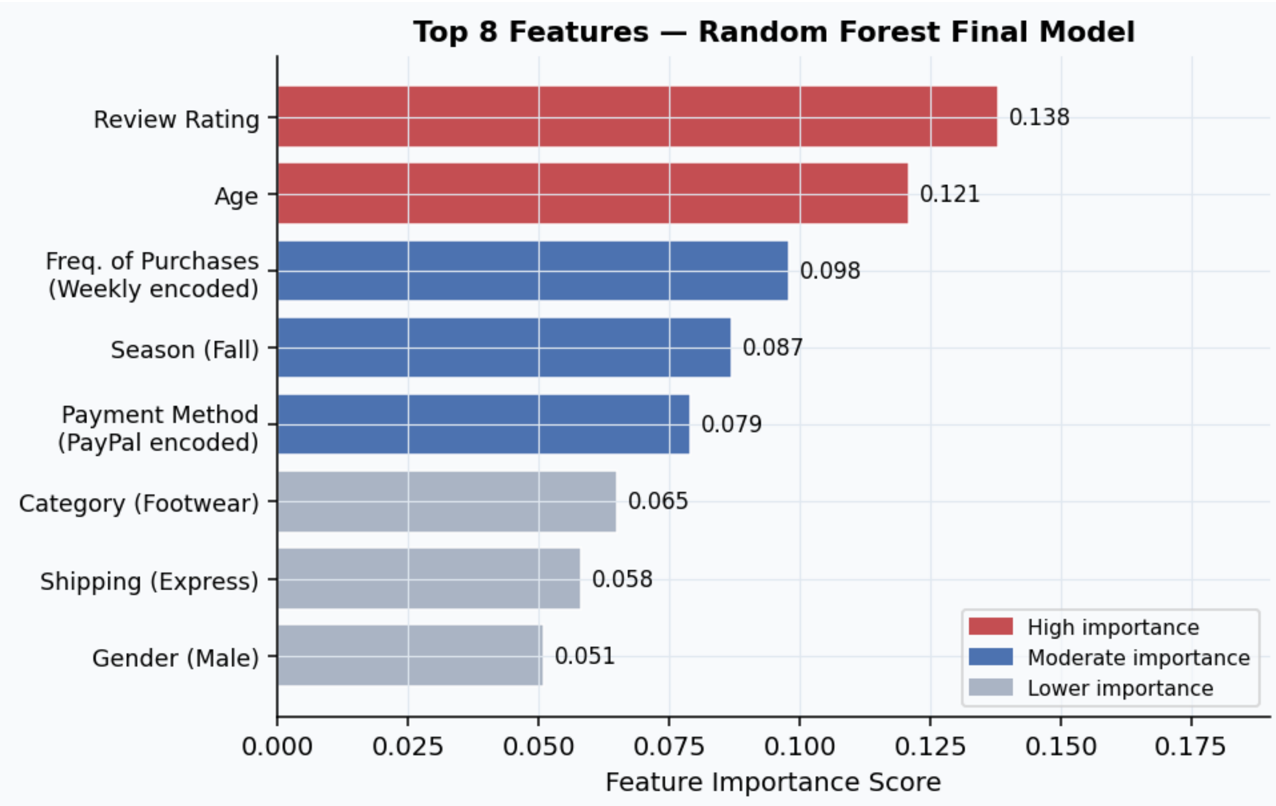

Figure 6 — Top 8 Feature Importances for the Random Forest final model. Review Rating and Age lead despite low linear correlation with the target, demonstrating the Random Forest's ability to capture non-linear interactions.

The top features confirm and extend the EDA findings: while Review Rating and Age showed weak linear correlation with the value tier, the Random Forest captures non-linear interactions that make them collectively predictive. Shopping frequency encodings and payment method also surface as meaningful signals in the ensemble.

7. Recommendations

Based on the model outputs and EDA findings, three concrete recommendations are offered to BigMart's marketing team:

Recommendation 1 — Deploy Tiered Marketing Campaigns

Deploy the Random Forest model to score the current customer database and create three distinct marketing tracks:

- **High-tier customers (★ Retain):** VIP loyalty rewards, early-access promotions, and personalised offers. The cost of losing a high-value customer far exceeds the cost of a retention incentive.
- **Medium-tier customers (↑ Convert):** Subscription trial offers and targeted discounts designed to move them into the High tier. EDA shows that converting a non-subscriber to a subscriber is the single highest-leverage action available.
- **Low-tier customers (■ Re-engage):** Low-cost email drip campaigns and seasonal promotions. Avoid heavy spend on this segment until behaviour signals an upward shift.

Target: All customers | **Expected impact:** More efficient budget allocation; reduced spend on low-ROI customers.

Recommendation 2 — Prioritise Subscription Conversion

Subscription status is the strongest single differentiator between High and Low value tiers (88% vs. 5% subscription rate). A targeted campaign offering free trials or first-month discounts to non-subscribed Medium-tier customers represents the most direct path to increasing the proportion of High-value customers in the base.

The model's feature importances also confirm that shopping frequency — closely linked to subscription engagement — is among the top predictors, reinforcing that subscription-driven behaviour, not just the subscription flag itself, elevates customer value.

Target: Medium-tier, non-subscribed customers | **Expected impact:** Increased High-tier volume; improved long-term revenue retention.

Recommendation 3 — Retrain the Model Quarterly

The current proxy target (Customer Value Score) was built from a static data snapshot. As new transaction data accumulates, the score distributions and tier boundaries will shift. Retraining the Random Forest quarterly — or whenever a significant volume of new customers is added — will keep tier assignments accurate and ensure the marketing strategy remains aligned with actual customer behaviour.

If BigMart can eventually capture true Customer Lifetime Value (CLV) data, the model should be retrained on that grounded label, replacing the proxy score entirely and significantly improving both accuracy and executive buy-in.

Target: Data / ML team | **Expected impact:** Sustained model accuracy and relevance of tier assignments over time.

8. Ideas for Further Research

- **Incorporate true CLV data.** If BigMart captures actual 12-month revenue per customer, the model can be retrained on a grounded label, replacing the proxy score and improving accuracy.
- **Explore gradient boosted models.** XGBoost or LightGBM typically outperform Random Forest on tabular datasets and support additional hyperparameter tuning via cross-validation.
- **Apply RFM segmentation.** If transaction timestamps become available, Recency-Frequency-Monetary scores would add a recency dimension absent from the current dataset.
- **A/B test marketing interventions.** A controlled experiment (withholding the campaign from a randomly selected control group) would isolate the causal lift from the segmentation strategy vs. the treatment alone.
- **Hyperparameter tuning.** Grid search or Bayesian optimisation over max_depth, n_estimators, min_samples_split, and class_weight could yield measurable F1 improvements.

9. Methodology Summary

Step	Action	Key Output
1. Problem Definition	Defined business need; formulated as multiclass classification	Problem Statement Worksheet
2. Data Collection	Loaded Kaggle Shopping Trends Dataset (3,900 rows, 19 cols)	Raw CSV
3. Data Wrangling	Verified zero missings/duplicates; engineered Customer Value Score + tier label	Cleaned CSV
4. EDA	Histograms, bar charts, heatmap; identified top 3 drivers; 6 figures produced	EDA Notebook
5. Preprocessing	Dropped leakage columns; one-hot encoded; 80/20 stratified split; StandardScaler	Preprocessed CSV
6. Modeling	Trained Logistic Regression, Decision Tree (max_depth=5), Random Forest (100 trees)	Model comparison table
7. Model Selection	Selected Random Forest (highest Macro F1); inspected feature importances	Final model
8. Documentation	Final report PDF, model metrics CSV, cleaned notebooks, GitHub repo	This document

Table 4 — Data Science Method (DSM) Step Summary

All code is available as Jupyter Notebooks in the project GitHub repository: Capstone3_Data_Wrangling.ipynb · Capstone3_EDA.ipynb · Capstone3_Preprocessing_Modeling_new.ipynb